

GDPS METRO GLOBAL – GM OVERVIEW



Agenda

- What
- Why
- Topology
- Technologies
- Tools
- Overview of scenarios

Just what is GDPS Metro Global - GM

- GDPS Metro Global – GM is also more simply referred to as GDPS MGM
- GDPS MGM is the combination of:
 - GDPS Metro
 - GDPS Global – GM

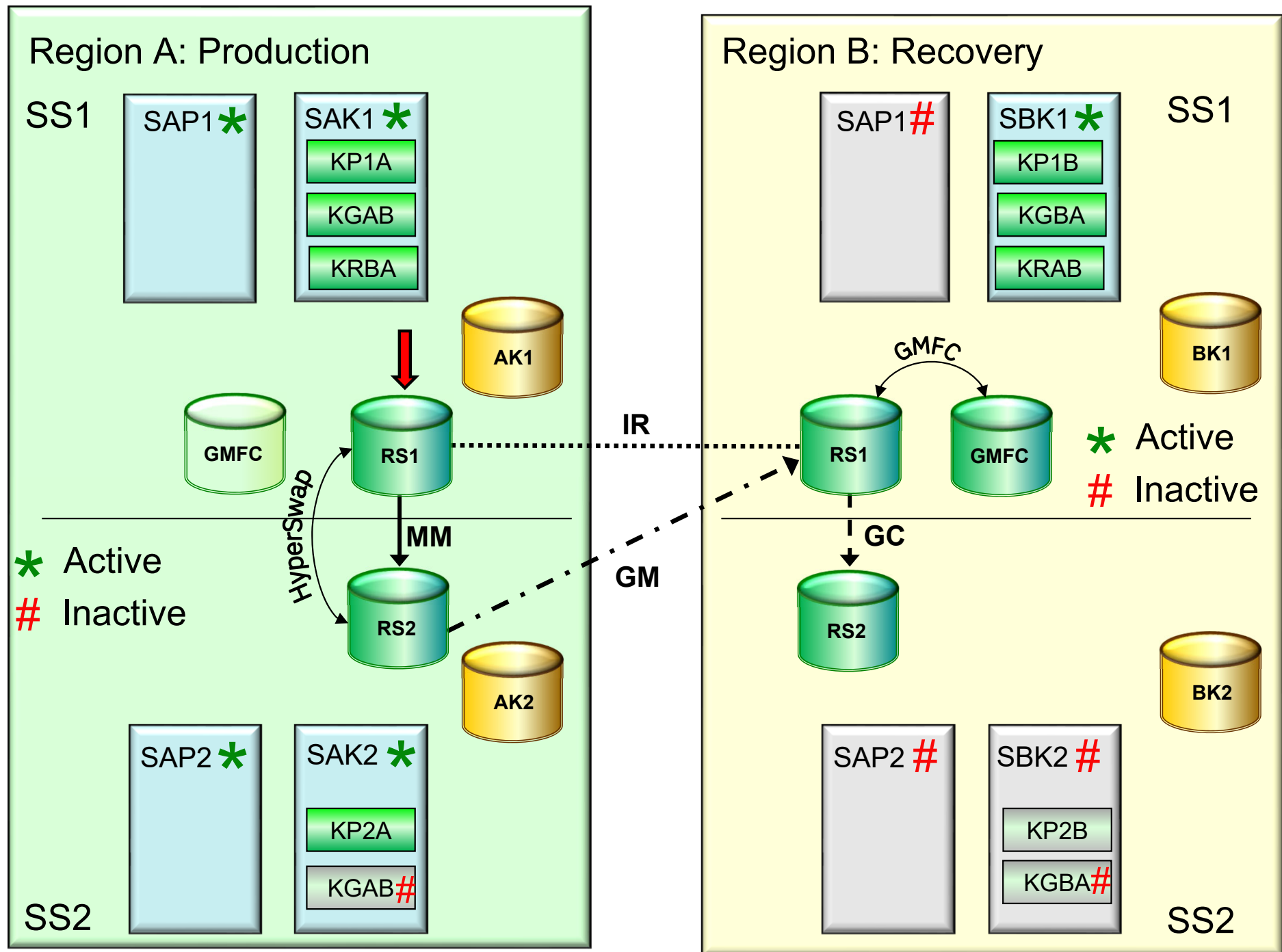
Business and regulatory pressures driving 4-site solutions

- Various financial services regulatory bodies are requiring long distance disaster recovery solutions to protect against wide-scale disruptions (regional disasters)
- Many customers especially in financial services already have short distance or campus synchronous solutions providing high/continuous availability and some measure of DR protection
- Business units are not willing to forego the availability benefits of synchronous mirroring solutions but need to implement longer distance recovery solutions
- In some cases the regulatory solution is seen as having a very low probability of being used and so objective is to satisfy regulator with minimum cost
- Other bodies, for example Governments, not wanting to carry blame for issues caused by outages setting requirements for zero outages

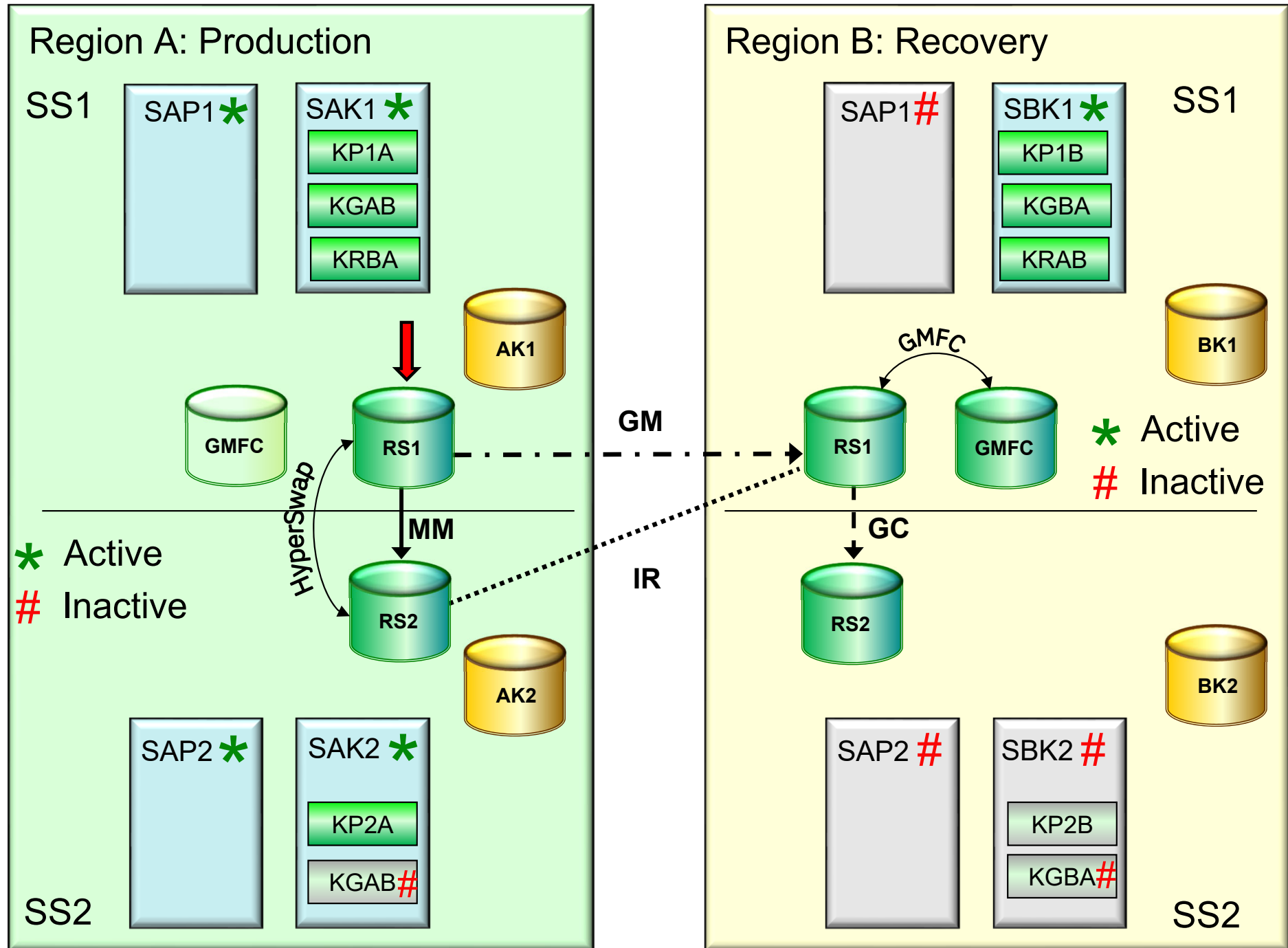
GDPS Metro Global GM Topology (4-site)

- Symmetrical disk replication combining multiple copy technologies
- The MGM 4-site topology could run either in a CASCADED or a MULTI TARGET configuration
- Each configuration has its benefits in certain scenarios
- Incremental Resynch between Sites and Regions is supported in both configurations
- Additional sets of disks for disaster validation and test are supported

GDPS MGM 4-site Topology (cascaded)



GDPS MGM 4-site Topology (multi-target)



Terminology

- Region
 - 2 physical regions
 - Production region:
 - usual location of the production systems and the primary disks
 - Alternate location of the GM secondaries and GM flash copy devices
 - Recovery region:
 - usual location of the GM secondaries and GM flash copy devices
 - Alternate location for production region and primary disks
 - Each region can have the role of the
 - Application region or
 - Recovery region
 - Terms used here:
 - Region A (production region)
 - Region B (recovery region)
- Server Site
 - Sites, where CPU resources are located
 - Typically Site1 (or SS1) and Site2 (SS2) in a GDPS Metro configuration
- Replication Site
 - Sites, where disk resources are located
 - Typically RS1 and RS2
 - Combined with region name: *region_name.RSx*

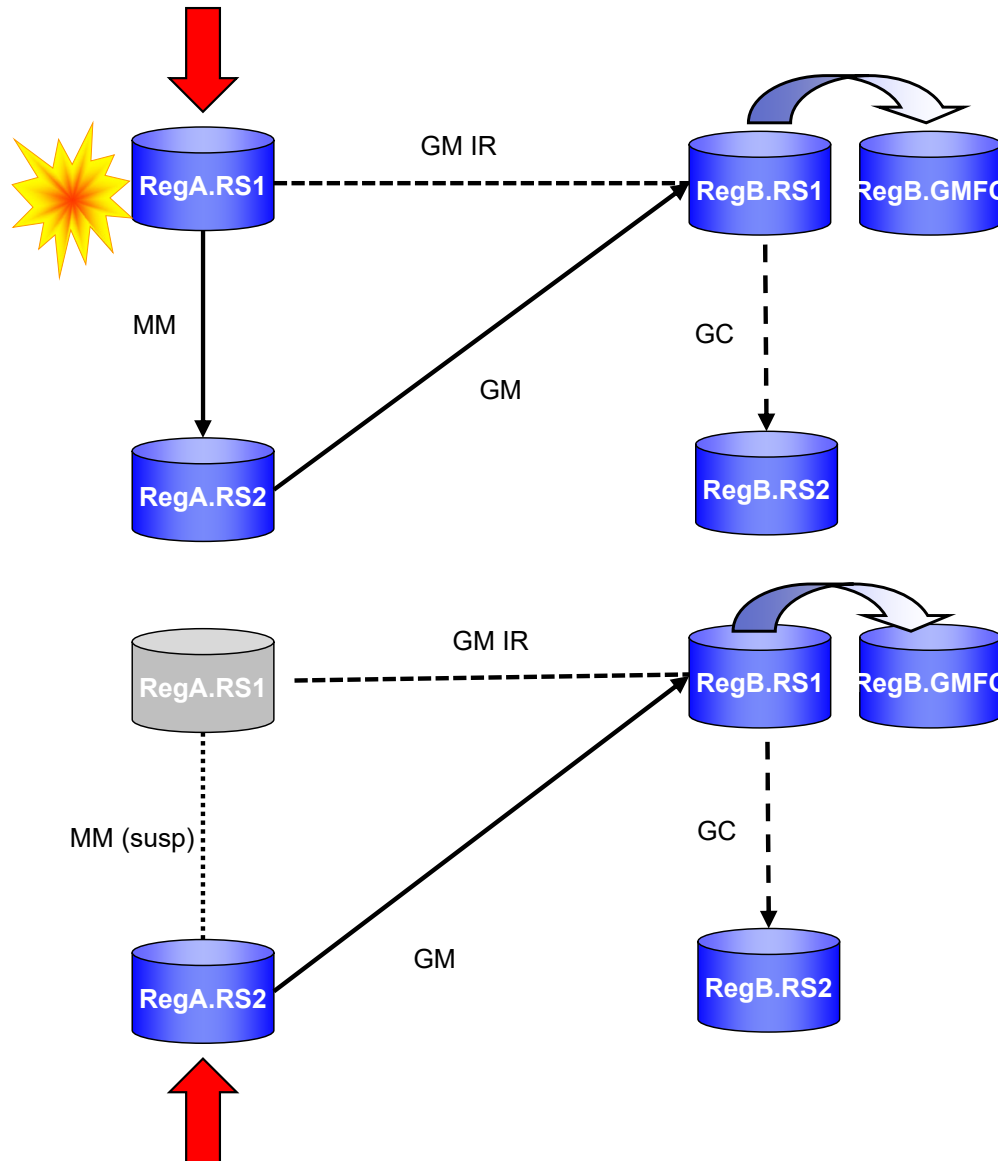
Terminology (cont.)

- Replication leg
 - Replication connection between disk locations with fixed names in a MGM configuration
 - Intra region leg: *region_name*.RL1 - the leg is located in region *region_name*
 - Cross region leg: *region_name*.RLxy - the leg has its source disk in region *region_name*, replication site x and target disk in replication site y
- Disk resources
 - Identified by *region_name*.RSx
 - Located in region *region_name* and replication site x
 - GMFC:
 - GM FlashCopy target to provide GM consistency
 - Identified by *region_name*.GMFC, where *region_name* specifies region where the disk are located

Terminology: System names

- Controlling system
 - Actually the controlling system function that runs in a NetView instance and enables GDPS to manage the MGM environment
 - Multiple functions/NetViews can run in a single z/OS image
 - Kp-sys: GDPS Metro Controlling system
 - 2 Controlling systems recommended (one in each server site)
 - Kp-sys means the current master
 - Kpx denotes a specific controlling system
 - Kg-sys: GDPS GM controlling system, located in both regions
 - Runs either in Site1 in a Cascaded or in Site2 in a Multi-target configuration in one of the Kp-sys z/OS images
 - a procedure to move Kg-sys between RS1 and RS2 should be in place
 - Kr-sys: GDPS GM Controlling systems in both regions
- Production system
 - System on behalf of which GDPS manages the remote copy functions
 - Not a Controlling system
- Recovery system
 - Production system running in a backup partition

Cascading vs. Multi Target consideration



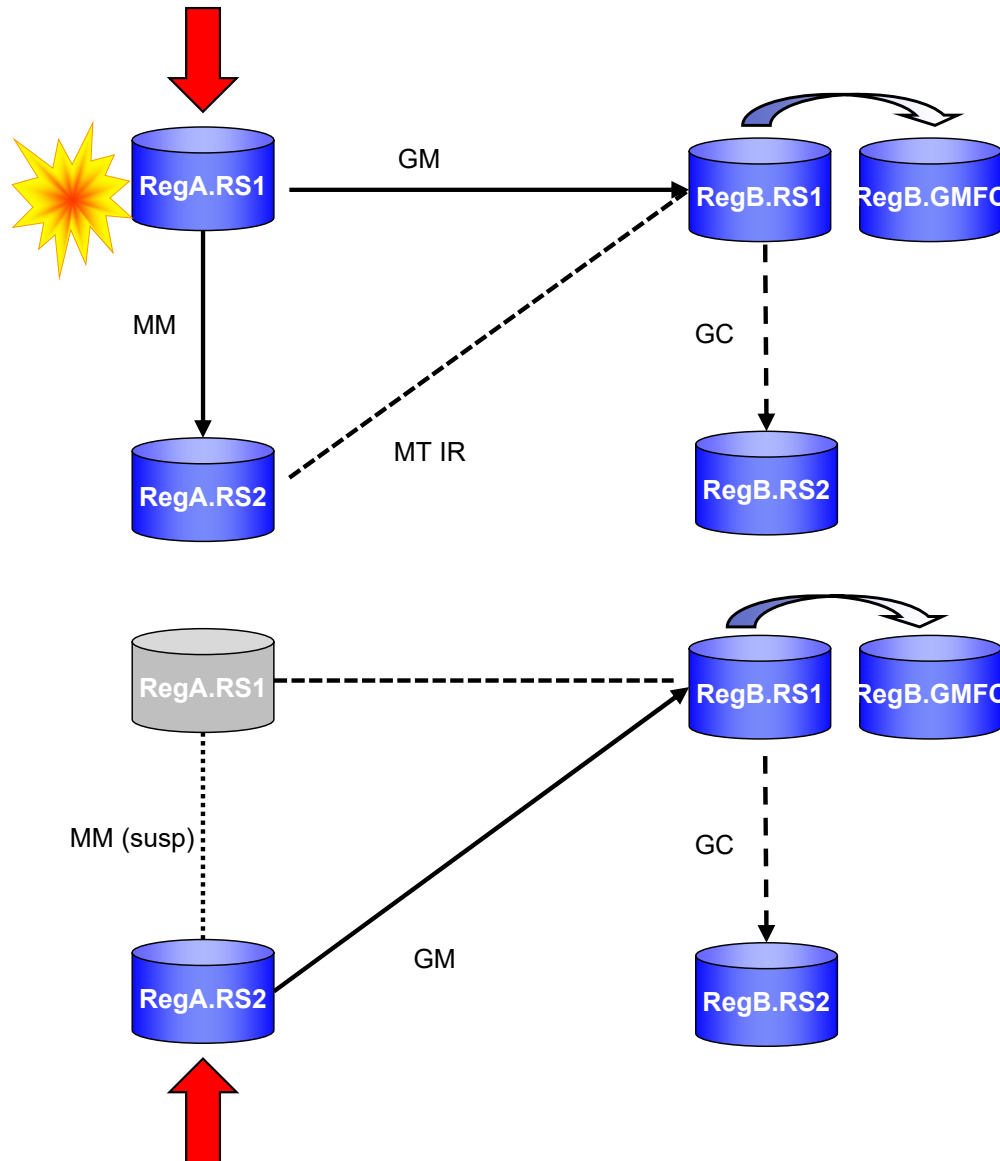
- Loss of RegA.RS1 in Cascaded config

- HyperSwap to RegA.RS2
- GM continues
- GC continues B.RS1 – B.RS2

- Reintroduction of RegA.RS1

- Resynch RegA.RS2>RegA.RS1
- HyperSwap to RegA.RS1
- Resynch RegA.RS1>RegA.RS2
- GC continues B.RS1 to B.RS2

Cascading vs. Multi Target consideration (cont.)



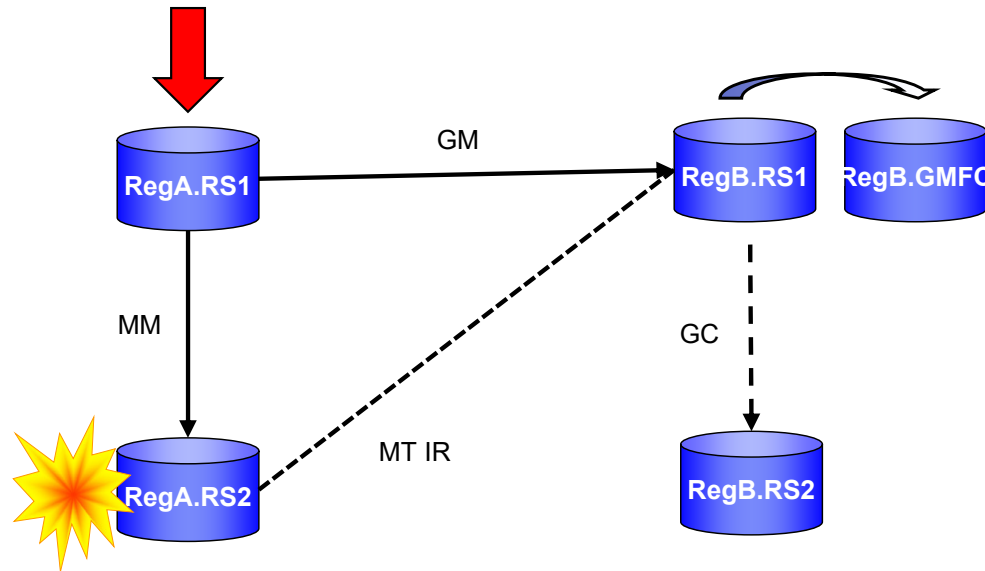
■ Loss of RegA.RS1 in a Multi-target config

- HyperSwap to RegA.RS2
- Move GM to RegA.RS2>RegB.RS1 (IR-supported)
- GC continues B.RS1 – B.RS2

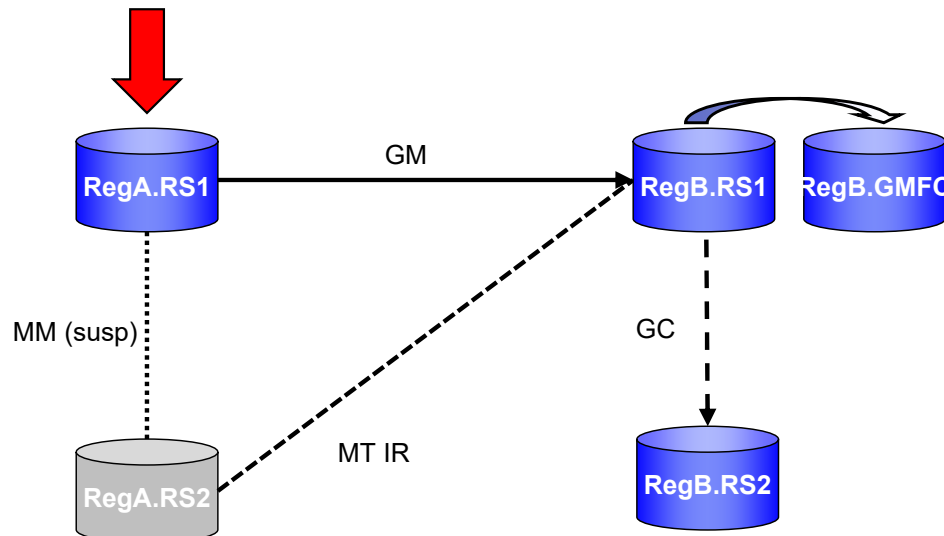
■ Reintroduction of RegA.RS1

- Resynch RegA.RS2>RegA.RS1
- HyperSwap to RegA.RS1
- Resynch RegA.RS1>RegA.RS2
- Move GM to RegA.RS1>RegB.RS1 (IR-supported)
- GC continues B.RS1 – B.RS2

Cascading vs. Multi Target consideration (cont.)

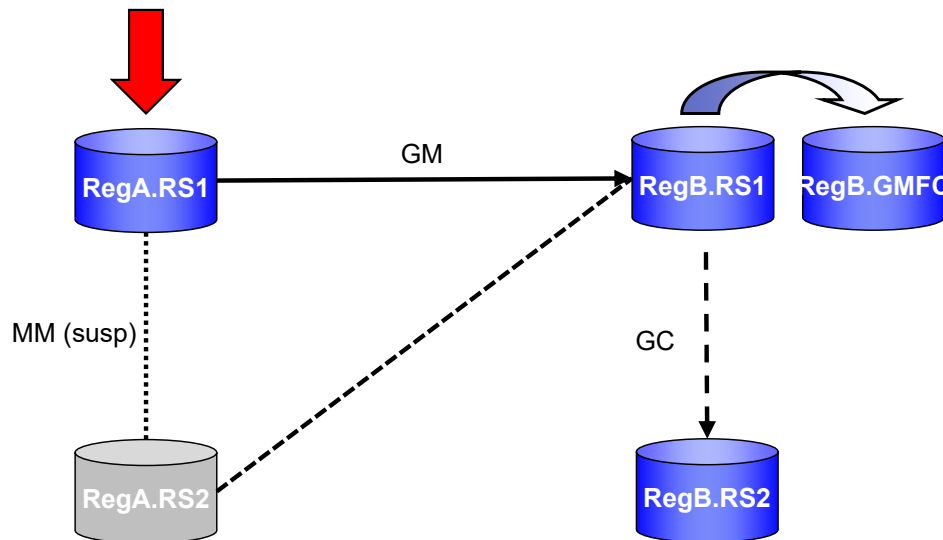
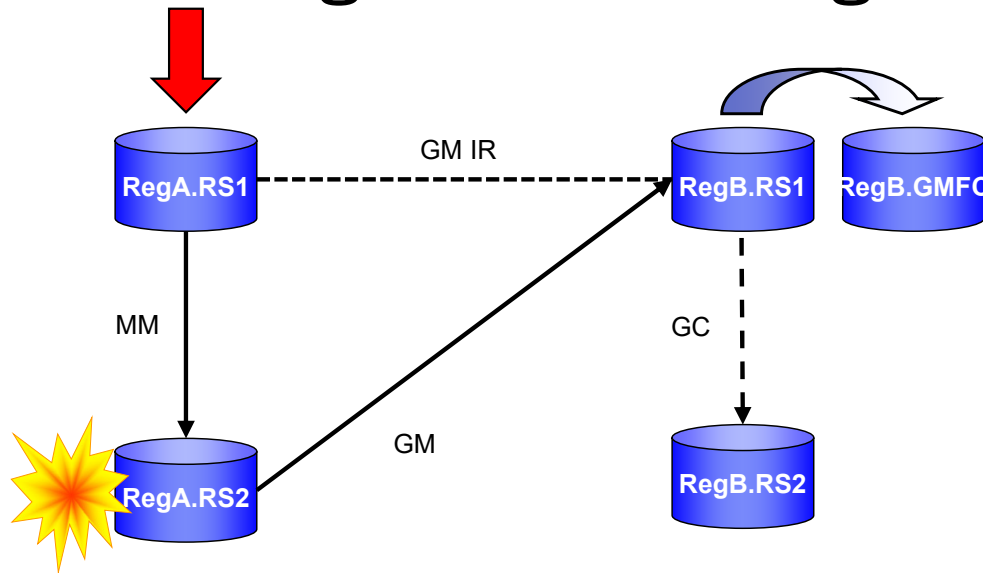


- Loss of RegA.RS2 in Multi-target config
 - Freeze/Suspend of RegA.RS2
 - GM continues
 - GC continues B.RS1 – B.RS2



- Reintroduction of RegA.RS2
 - Resynch RegA.RS1>RegA.RS2
 - GC continues B.RS1 – B.RS2

Cascading vs. Multi Target consideration (cont.)

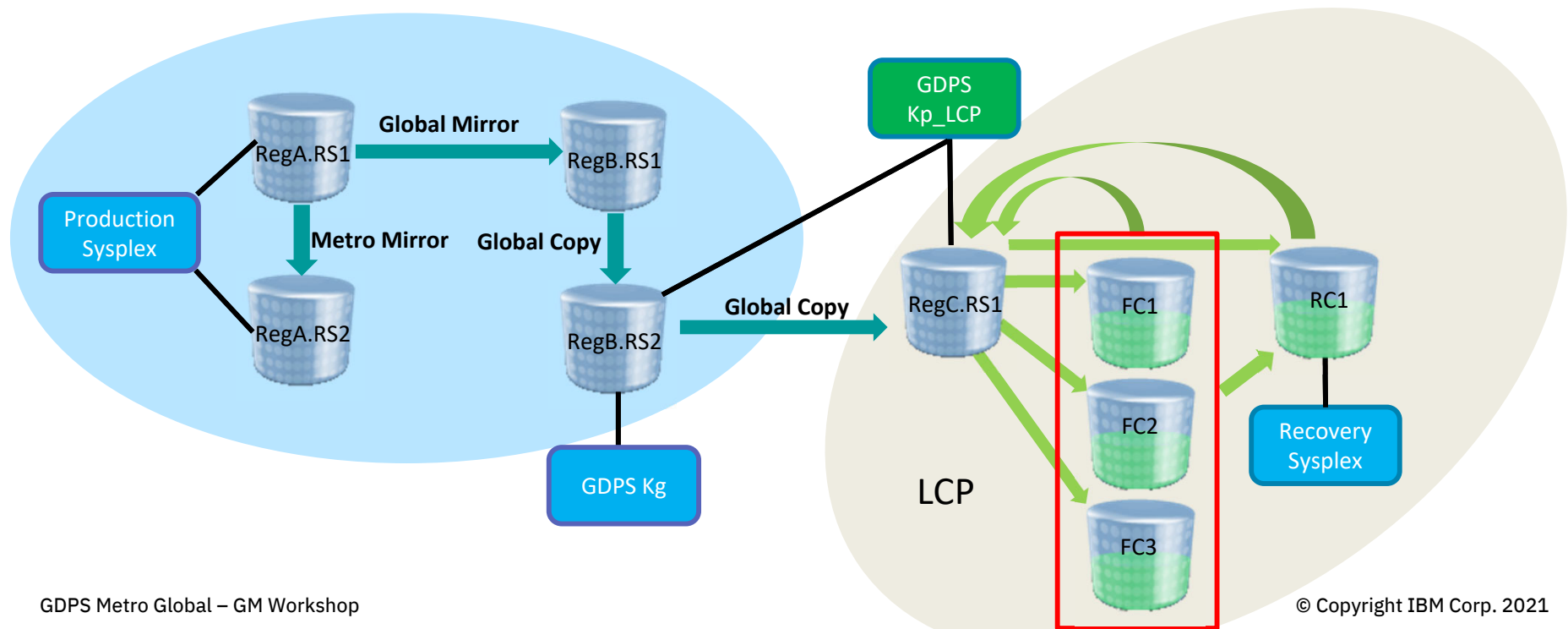


- Loss of RegA.RS2 in Cascaded config
 - Freeze / Suspend of RegA.RS2
 - Move GM to RegA.RS1>RegB.RS1 (IR-supported)
 - GC continues B.RS1 – B.RS2

- Reintroduction of RegA.RS2
 - Resynch RegA.RS1>RegA.RS2
 - Move GM to RegA.RS2>RegB.RS1 (IR-supported)
 - GC continues B.RS1 – B.RS2
- There is no general benefit of one configuration over the other
- Each configuration has its advantages in certain scenarios

Logical Corruption Protection – 4-site

- External LCP and Testcopy Manager can be combined with GDPS MetroGlobal GM 4-site by a cascaded Global Copy
- LCP provides protection against cyber attacks, internal threats and other forms of logical corruption
- Testcopy Manager allows to manage, capture, and refresh a test copy for use with a separate, isolated test environment
- An additional GDPS controlling system is required



Technologies used with MGM4SITE

- Incremental Resynchronization (IR) :
 - Capability to only copy those data between the remaining replication sites that have changed since the loss of a replication site
- Remote Script Execution (RSE) :
 - Capability to initiate actions in different GDPS environments that form the overall solution from a Single Point of Control
- Procedure Handler:
 - with provided procedures the procedure handler can drive complex scenarios with a single script statement
 - makes use of RSE

Incremental Resynchronization (IR)

- Important capability to restore DR readiness immediately after loss of a replication site by only copying those data between the remaining replication sites that have changed since the event of disruption
- In cascaded configuration:
 - Loss of intermediate replication site RegA.RS2
 - GM session to be moved from RegA.RS2>RegB.RS1 to RegA.RS1>RegB.RS1
 - Only data created since loss of RegA.RS2 will be copied from RegA.RS1 to RegB.RS1 (no initial copy necessary)
- In multi-target configuration:
 - Loss of common source replication site RegA.RS1
 - GM session to be moved from RegA.RS1>RegB.RS1 to RegA.RS2>RegB.RS1
 - Only data created since loss of RegA.RS1 will be copied from RegA.RS2 to RegB.RS1 (no initial copy necessary)

Remote Script Execution (RSE)

- Capability to initiate actions in different GDPS environments that form the overall solution from a Single Point of Control
- Exploited in individual scripts and by the Procedure Handler
- Terminology:
 - A local script defined in the local initiator (GDPS system that initiates the script), calls a remote script, defined in the remote target (GDPS system that is the target of a RSE request) and waits for the completion of the entire remote script before it continues with the next script statement
 - Also a remote target group consisting of remote GDPS systems can be specified by a remote script statement: the local initiator will determine which system in the group is currently the appropriate system to execute the script
- Enabled by communication pathes between all Controlling systems:
 - Based on TCP/IP communication with NetView RMTCMD
 - Defined in GEOGROUP member (XML) allocated by NetView procedure

Procedure Handler

- An integrated environment to execute GDPS supplied procedures
- These procedures drive complex reconfiguration scenarios
- Invoked by the PROCEDURE *procname* script statement
- Follows a standard naming convention:
topology_action_currentconfig_currentreplicationsite_targetreplicationsite
- All procedures must be run in the Kg-sys

Alternate Subchannel Set support

- Exploiting the combined alternate subchannel set support of GDPS Metro and GDPS GM:
 - GDPS Metro allows defining the Metro Mirror secondary devices in MSS1
 - GDPS Global GM allows
 - defining the GM FlashCopy target devices in alternate MSS1 in the recovery site (or now on GDPS 4.4 with NO UCB)
 - not defining the practice FlashCopy target devices at all to the GDPS GM R-sys
 - GM primary devices in MSS1 (what are in MGM the MM secondary devices, Kp-sys and Kg-sys use same IODF of same z/OS image)

Tools

- Tools provide complementary function to GDPS
 - made available on an ‘as is’ basis
 - provided in source code format to allow modification to suit client needs
- GDPS EasyLog Tool
 - Microsoft Windows-based tool to help you extract and easily download the MVS Syslog and NetView log from a z/OS environment
 - helps in analyzing the Netlog after it is downloaded to a workstation
- RESTful API
 - API that uses HTTP or HTTPS requests to GET or POST data to GDPS to make requests to obtain information or perform actions
 - is based on REpresentational State Transfer (REST) technology

Tools (cont.)

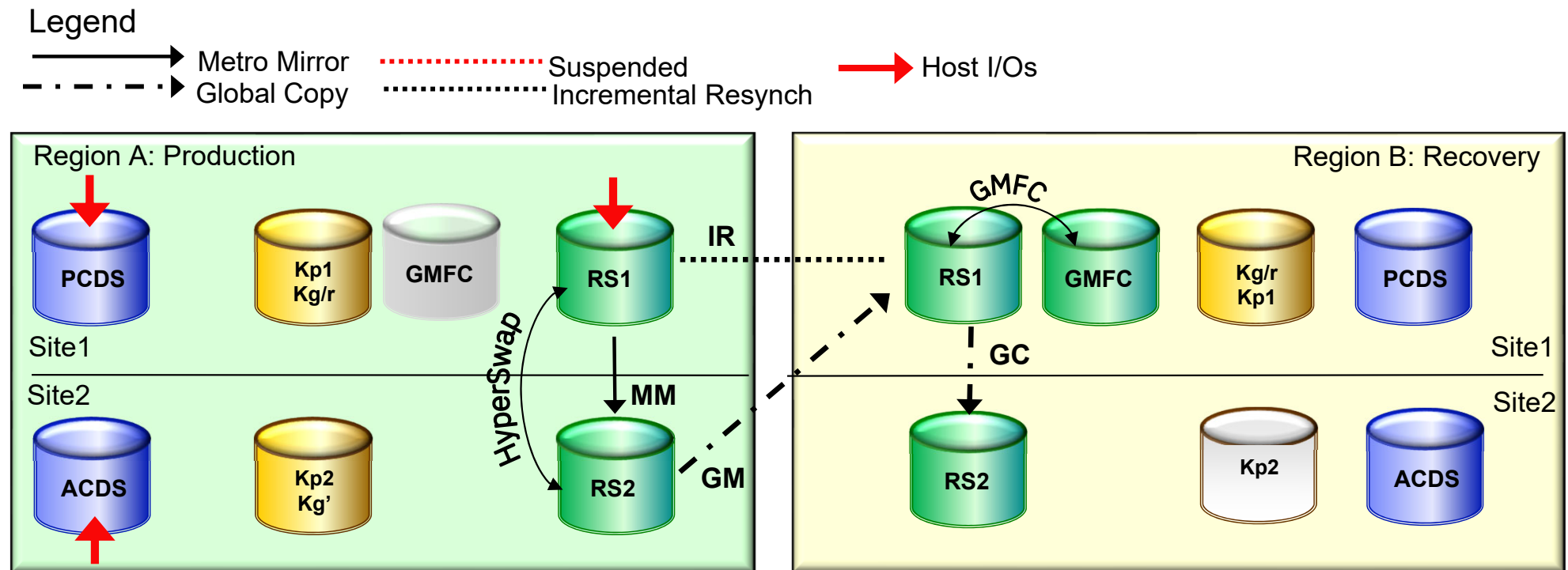
- Query Services
 - allows user-written REXX programs running under NetView to query and obtain the value for some GDPS variables
 - no additional installation tasks required
- GDPS GUI
 - Graphical representation of GDPS status and management capability
 - XML editor for managing configuration data

Overview of MGM scenarios

- Planned primary disk outage and return to normal
- Unplanned primary disk outage and return to normal
- Planned secondary (intermediate) disk outage and reintroducing secondary disk (Freeze and Resynch)
- Unplanned secondary (intermediate) disk outage (Freeze and Recover) and reintroducing secondary disk
- Planned region switch and return home
- Unplanned region switch and return home after a disaster
- Planned Recovery Testing on isolated disk

Notes

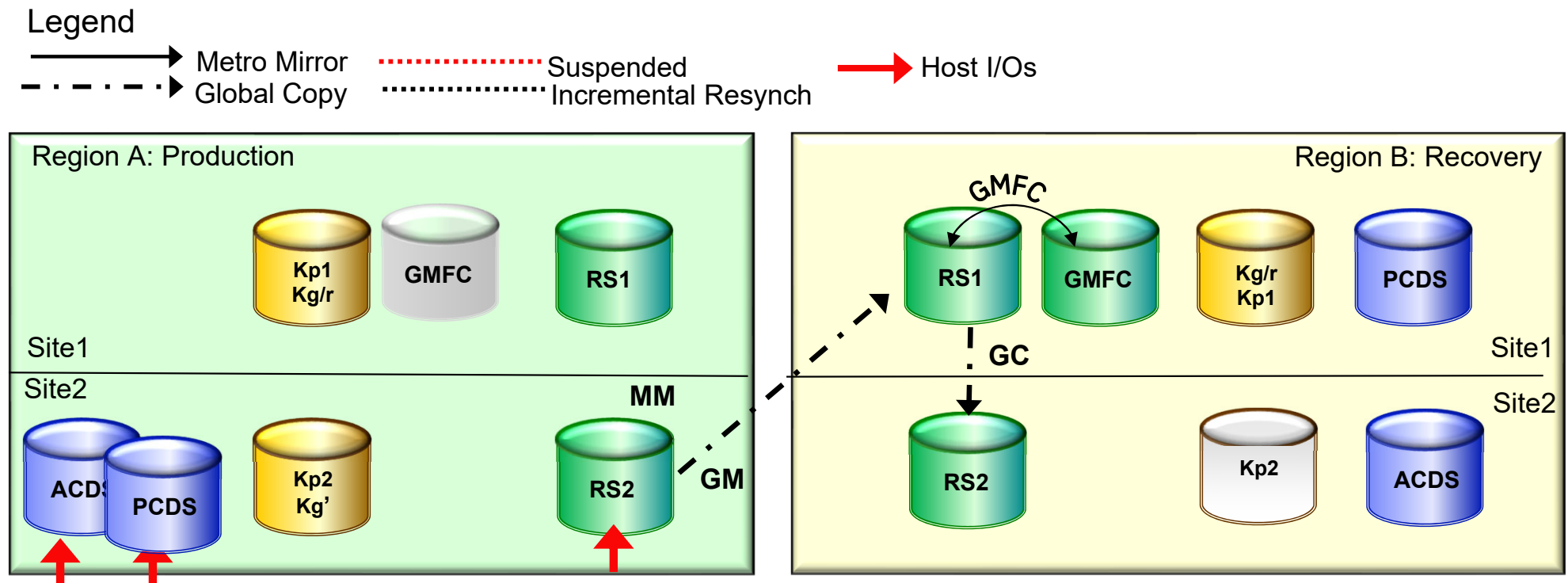
- Starting point is a standard MGM 4-site cascaded configuration



- All outage scenarios assume that disks that incur the outage are returned to the configuration 'intact'
 - Data on the disk is intact
 - Copy services relationships are intact

Unplanned primary disk outage

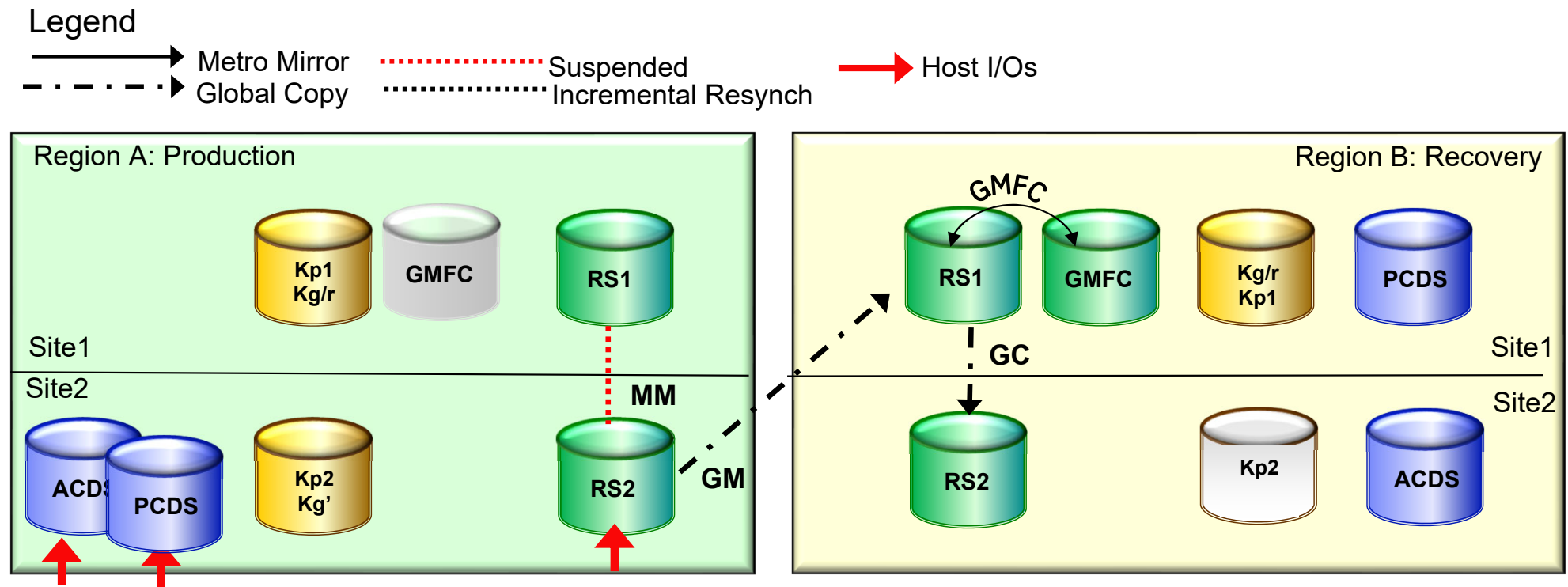
■ Configuration after primary disk outage



- After HyperSwap RegA.RS2 are Primary Suspended(3)
- CDS isolated in Site2
- GM continues
- GC continues in Region B
- If Site1 simplex disk are lost: Kp2 stays Master, Kg-sys moved to Site2 by own procedure

Planned primary disk outage

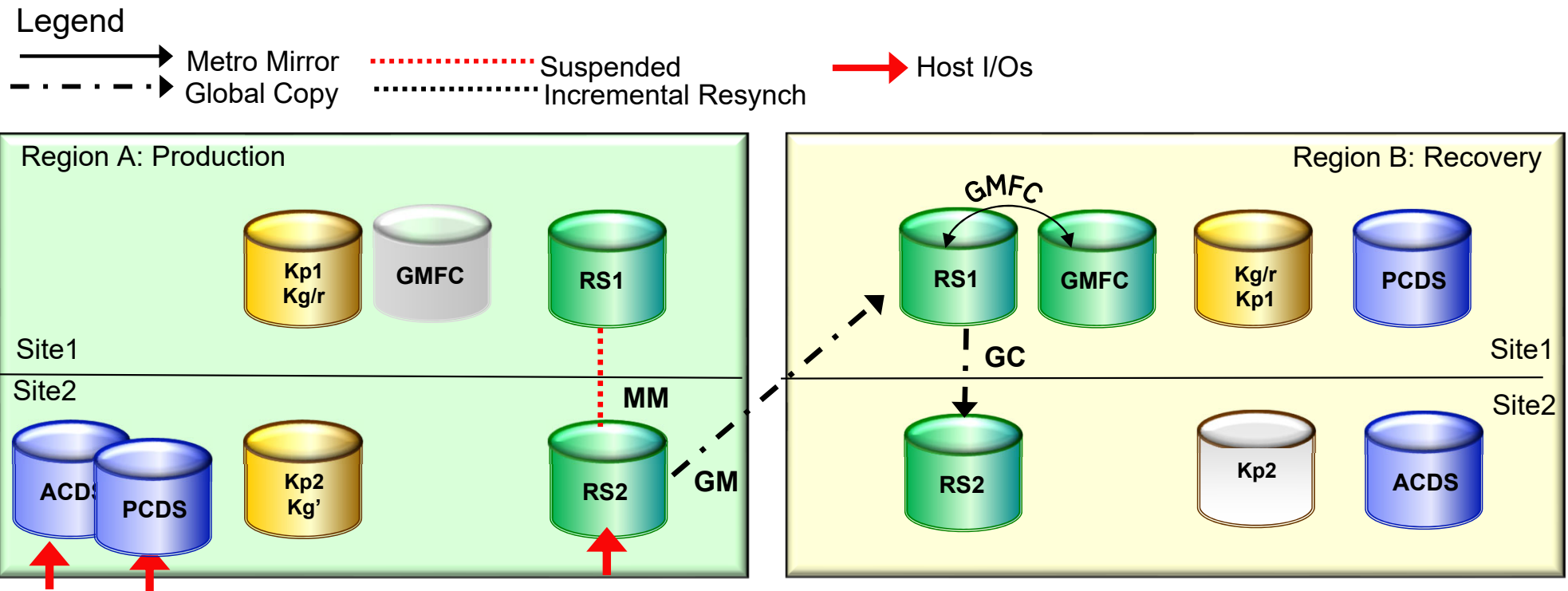
- Configuration after primary disk outage (due to maintenance)



- Kg-sys moved to Site2 by own procedure
- Planned HyperSwap: RegA.RS2 are Primary Suspended(3)
- CDS isolated in Site2
- GM continues
- GC Continues in Region B

Return to normal after primary disk outage

- Goal: maintain HA and DR readiness as much as possible
- Configuration before returning to normal:

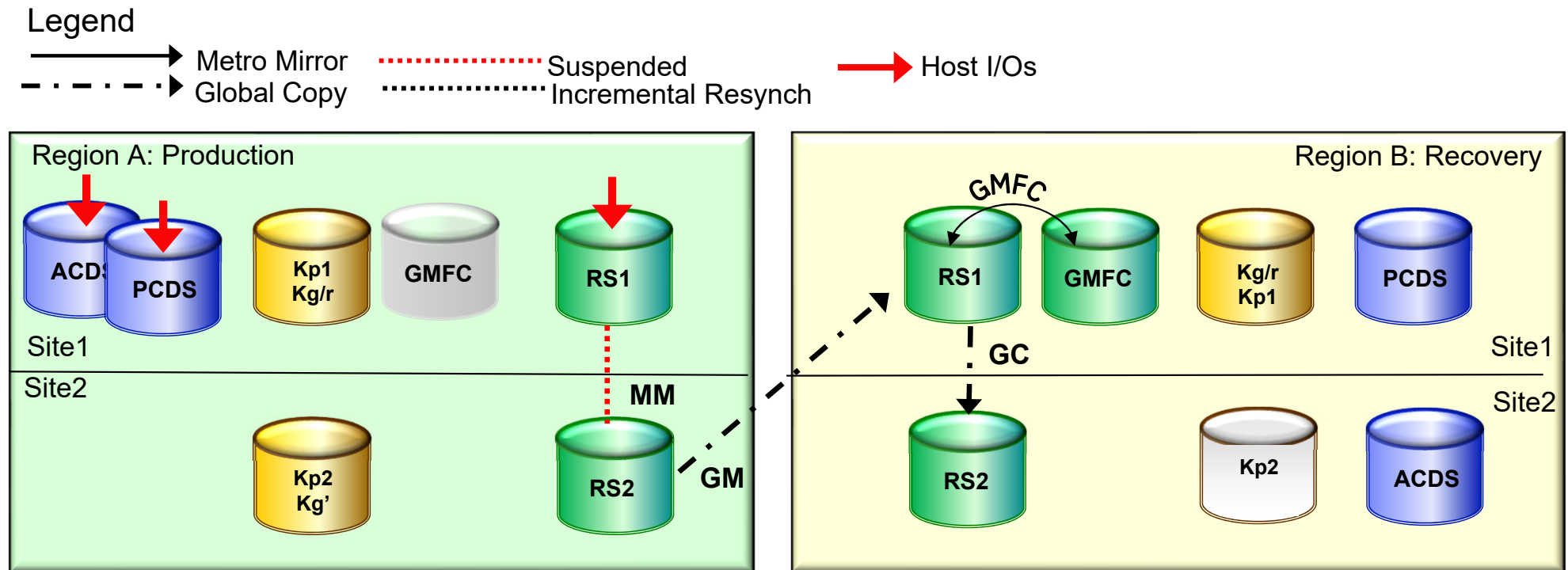


Steps to perform

- Move Kg-sys to Site1 by own procedure
- Establish Metro Mirror from RegA.RS2 to RegA.RS1 (ends up in MT config)
- Planned HS to RegA.RS1 + establish Metro Mirror to RegA.RS2 (GM ages) vs. procedures MGM4SITE_RESTORE + MGM4SITE_SWAP

Planned/unplanned secondary disk outage

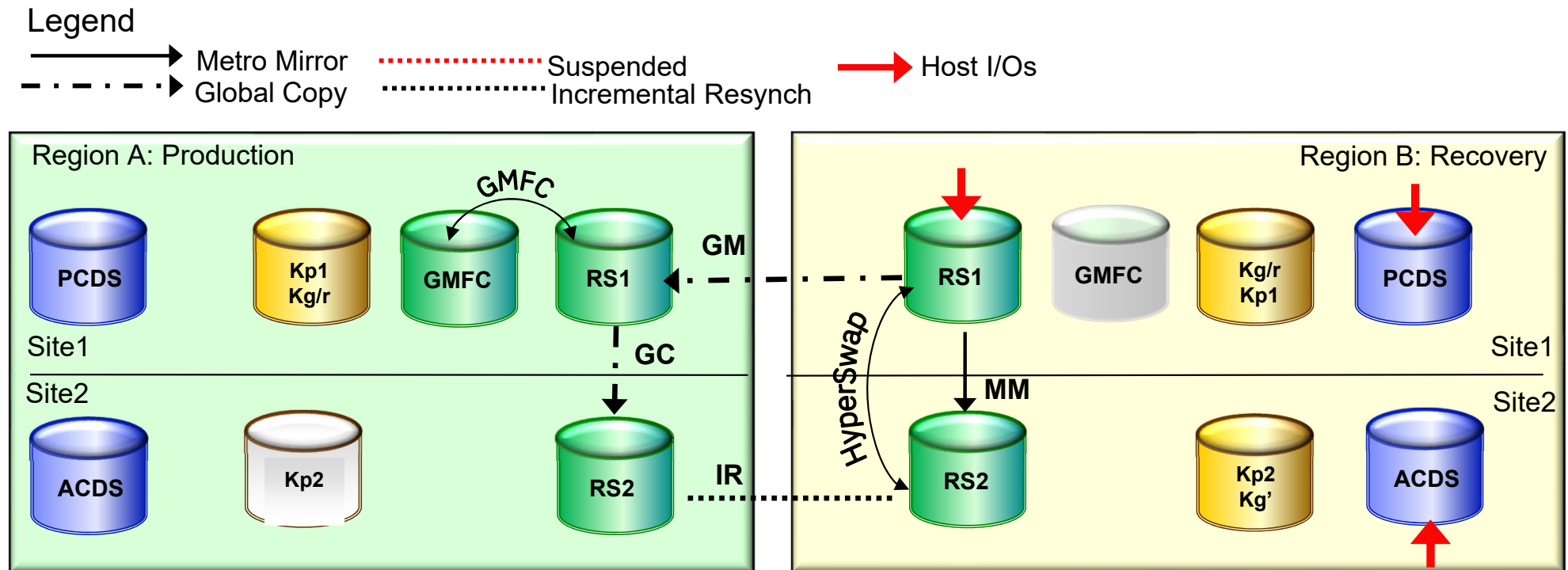
- Configuration after secondary disk outage



- RegA.RS2 frozen
- CDS isolated in Site1
- GM moved to RegA.RS1-RegB.RS1

Planned Region Switch

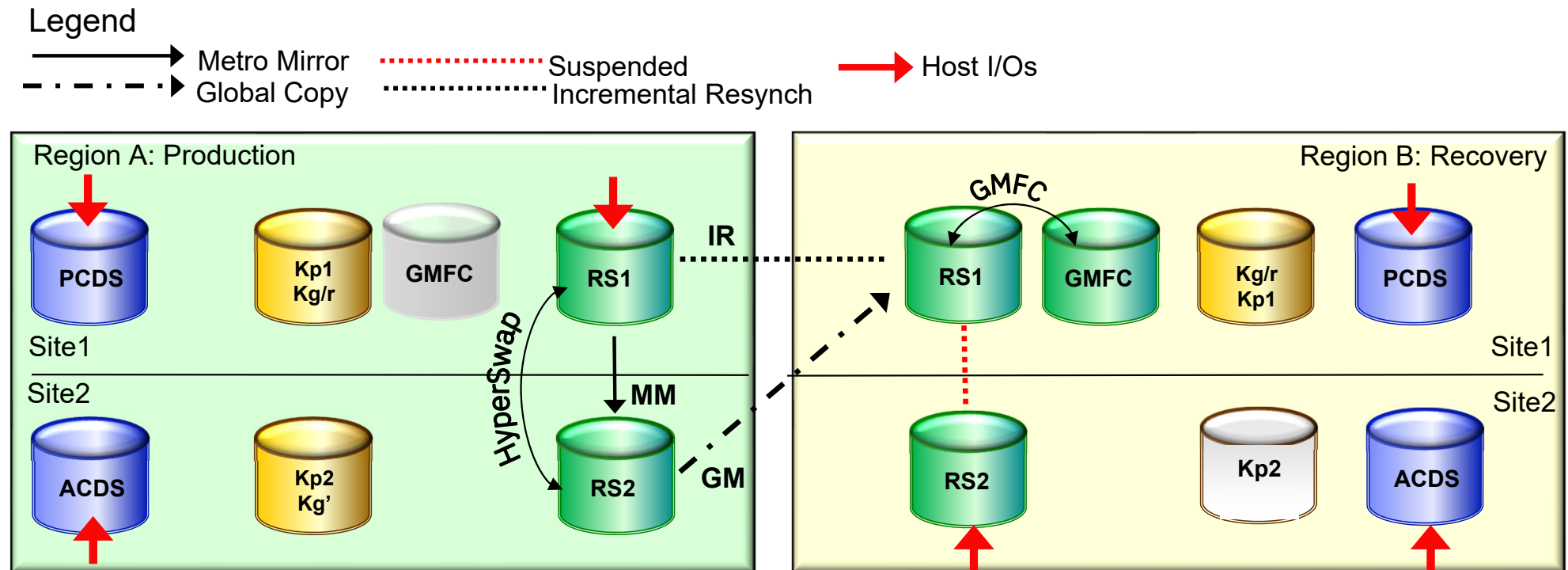
■ Configuration after region switch



- RegB.RS1 recovered
- Production running from RegB.RS1
- Mirror reconfigured, running with 3 copies cascaded: RegB.RS1 to RegA.RS1 to RegA.RS2

Planned DR Test

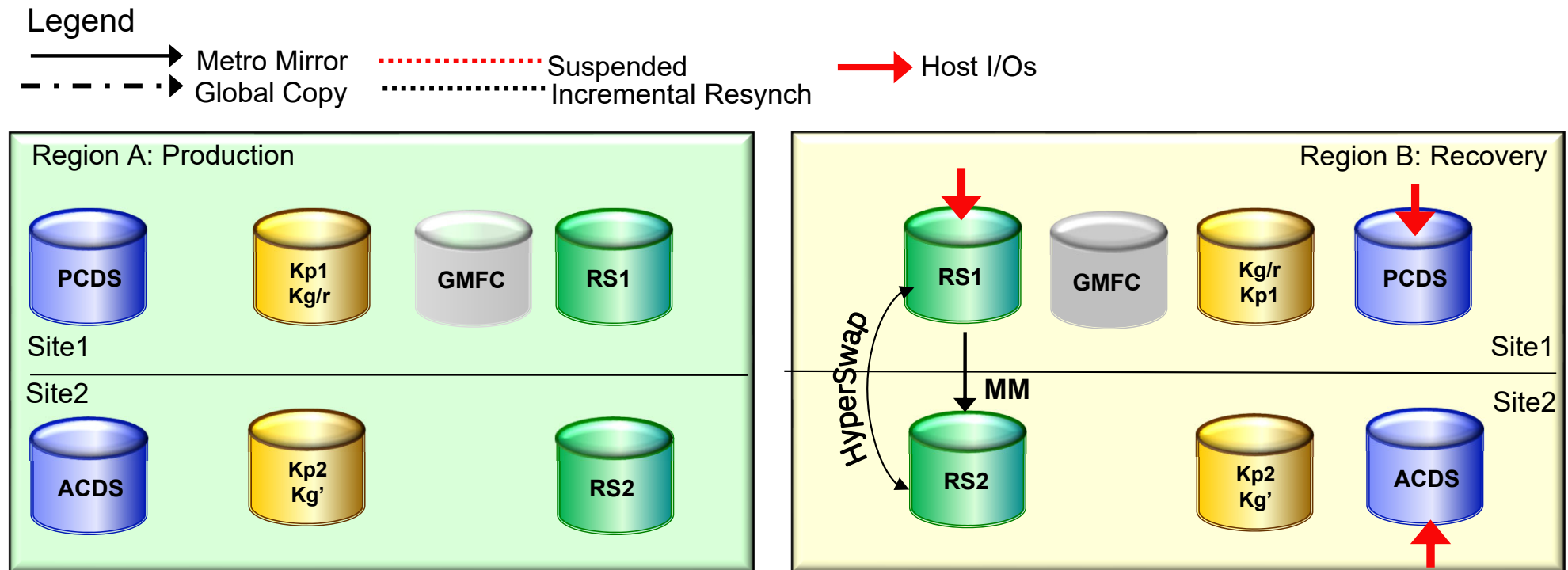
■ Configuration during DR Testing



- RegB.RS1 failed over (accumulating updates)
- Production DR Testing running from RegB.RS2
- Hyperswap disabled to RegB.RS2

Disaster Recovery

- Configuration after unplanned DR switch



- RegB.RS1 recovered
- Production running from RegB.RS1
- Hyperswap Enabled to RegB.RS2

END OF LECTURE